

# Project Schedule and Resource Planning

The following schedule outlines the planned development activities, estimated duration, project milestones, resource allocation, and potential risks associated with the Smart Lender Loan Approval Prediction System. The plan ensures systematic execution, efficient resource utilization, timely completion, and successful deployment of the proposed solution.

## 1. Project Timeline

|   |                                             | Estinuted Duration | <b>Deliverable</b>                                   |
|---|---------------------------------------------|--------------------|------------------------------------------------------|
| 1 | Requirement Analysis                        | 1 week             | Software Requirement Specification (SRS)             |
| 2 | data Collection and Analysis                | 1Week              | Dataset                                              |
| 3 | Data pre-processing and Feature Engineering | 1 Week             | Cleaned and Processed Data                           |
| 4 | Machine Learning Model Development          | 2 Weeks            | trained and Evaluated Prediction Model               |
| 5 | Flask Web Application Development           | 2 Weeks            | Functional Web Application                           |
| 6 | System Testing and Validation               | 1 week             | Testing and Validation Report                        |
| 7 | Deployment and Documentation                | 1 Week             | Deployed Application and final Project Documentation |

| <b>Resource</b>            | <b>Technology</b>           | <b>Purpose</b>                                                                            |
|----------------------------|-----------------------------|-------------------------------------------------------------------------------------------|
| Programming Language       | Python 3.x                  | Machine Learning Model Development and backend programming.                               |
| Backend Framework          | Flask                       | REST endpoints, business logic, authentication, and prediction                            |
| Frontend                   | HTMLS, c.SS3, java script   | Builds a responsive and user friendly web interface,                                      |
| Database                   | MySQL                       | Stores applicant information, user accounts, application and prediction history securely. |
| Machine Learning Libraries | Scikit-learn, Pandas, NumPy | Data preprocessing, feature engineering, model training, and evaluation                   |

|                   |                                                             |                                                               |
|-------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| Development Tools | Visual Studio Code,<br>Jupyter Notebook,<br>Postman, Joblib | Coding, experimentation, model serialization, and API testing |
| Version Control   | Git and GitHub                                              | Source code management, version tracking, and collaboration   |

2. Major Milestone

| Milestone                      | Expected Outcome                                            |
|--------------------------------|-------------------------------------------------------------|
| Requirement Analysis Completed | Approved project requirements                               |
| Dataset Prepared               | Clean Dataset Ready for Training                            |
| Machine Learning Completed     | Best medic-lion Model Selected                              |
| Web Application Developed      | Prediction system successfully integrated                   |
| System Testing Completed       | Verified Functional System                                  |
| Deployment Completed           | Application Successfullly Hosed and Ready for Demonstration |

3. Risk Assessment

|                          |                                                               |
|--------------------------|---------------------------------------------------------------|
|                          | Mitigation                                                    |
| Poor Data Quality        | Perform thorough preprocessing, validation, and cleaning      |
| Low Prediction Accuracy  | Train multiple models and select the best-performing one.     |
| Deployment Failure       | Test deployment using containers before production.           |
| Security Vulnerabilities | Implement secure authentication and encrypted communication.  |
| Projedct Delyas          | Monitor milestones regularly and distribute tasks efficiently |

4. Resource Allocation

|        |  |                                                                                                                            |
|--------|--|----------------------------------------------------------------------------------------------------------------------------|
| Cloud  |  | Hosts the Flask web application with reliable infrastructure                                                               |
| Docker |  | Packages the application into portable containers for consistent development, testing, and deployment across environments. |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conclusion                                                                                                                                                                                                                                                                                                                                                 |
| The project planning strategy provides a structured roadmap for implementing the Smart Lender Loan Approval Prediction System. By defining clear phases, milestones, resource allocation, and risk mitigation strategies, the project ensures efficient development, accurate prediction performance, secure data handling, and reliable cloud deployment. |